

Project Executive Summary: Customer Churn Platform

End-to-End Predictive Pipeline & Live Web Application

Technology Stack: Python, Streamlit, Scikit-Learn, Pandas, NumPy, Machine Learning Serialization (Pickle)

1. PROJECT OBJECTIVE

The purpose of this project was to convert raw, unstructured e-commerce transaction history into a business intelligence solution. By processing customer buying frequencies, transaction recency, and overall financial spending values, the platform constructs an automated classification system capable of highlighting high-value accounts showing risk signals of defection (churn) before they stop purchasing.

2. TECHNICAL IMPLEMENTATIONS (WHAT I DID)

Phase	Core Execution Details
Data Engineering	Ingested and parsed customer transaction logs (<code>online_retail.csv</code>). Conducted records deduplication, isolated null identifiers, and computed cross-feature transformations on product volumes and individual prices.
RFM Feature Extraction	Aggregated raw event logs into structural client dimensions tracking exactly how long since a customer's last order (Recency), frequency of return purchases (Frequency), and overall absolute revenue generated (Monetary).
Machine Learning Training	Constructed an algorithmic feature-scaling and classification pipeline inside a Jupyter environment (<code>Deployment_project.ipynb</code>) to mathematically separate high-retention users from churn-likely accounts.
Application Deployment	Built a standalone front-end reporting web application interface (<code>app.py</code>) using the Streamlit framework, allowing operational staff to interact with user features and trigger cloud-inference scoring instantly.

3. CONCRETE DELIVERABLES (THE OUTPUT)

The successful execution of this architecture resulted in five specific, high-value technical assets:

- **app.py (The Interface):** A functional, web-accessible user interface featuring interactive parameter selection boxes, live graphical summaries, and manual customer evaluation panels.
- **churn_model.pkl (The Brain):** A fully trained and serialized binary classification model capable of generating real-time predictions based on customer behavioral data inputs.
- **scaler.pkl (The Translator):** A serialized feature normalization model mapping arbitrary user entries into precise algorithmic matrices without baseline distribution distortion.
- **online_retail.csv (The Feature Database):** A structured, tabular feature store organizing historical transaction metadata, customer groupings, and calculated risk factors.
- **requirements.txt (The Environment):** A production-ready dependency blueprint mapping every micro-library version required to instantly deploy this ecosystem into cloud containers or remote web servers.

4. CORE BUSINESS OUTCOMES

- **Proactive Churn Mitigation:** Shifts retention workflows from a reactive posture to a proactive data-driven strategy, allowing companies to detect slipping customers months in advance.
- **Precision Marketing Alignment:** Segment mapping exposes precisely where revenue is concentrated, allowing executive teams to allocate marketing capital efficiently toward high-value cohorts.